Customer Churn Prediction and Business Risk Analysis

An End-to-End Machine Learning System and Value-Based Retention Strategy

Prepared for: Advanced Machine Learning & Retention Risk Management Analysis
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1. Executive Summary

Customer churn represents the terminal point of an active customer relationship where a subscriber terminates or stops using the telecom company's services. Because acquiring new customers is significantly more expensive than retaining existing ones, preventing churn is a core revenue-protection strategy for the organization. This project establishes a rigorous, end-to-end predictive analytics framework on the Telco Customer Churn dataset to identify likely leavers early enough to deploy targeted, cost-effective retention campaigns.

Rather than simply classifying customers with a rigid binary label (i.e., 'will churn' vs. 'will not churn'), this machine learning system produces stable, calibrated churn probabilities. These continuous scores are essential for prioritizing retention activities and balancing limited budgets. The pipeline evaluated five candidate algorithms: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Gradient Boosting. Gradient Boosting was selected as the final champion model based on its training-set cross-validation performance, achieving a mean Area Under the Receiver Operating Characteristic (ROC-AUC) of 0.849. This performance was firmly validated on an untouched, 20% holdout test set with an ROC-AUC of 0.845, demonstrating strong generalization.

The key performance metrics and model specifications are summarized below:

Operational Dimension	Result Value	Strategic Significance
Final Classifier Model	Gradient Boosting	Ensemble model sequentially minimizing prediction loss
Selection Validation Basis	5-Fold CV ROC-AUC	Cross-validation ensures statistical model stability
Holdout ROC-AUC Score	0.845	Demonstrates excellent probability score separation
Holdout Classification Accuracy	80.4%	Overall percentage of correct predictions
Tuned Holdout Precision (Churn)	67.3%	Minimizes financial waste on loyal customers
Tuned Holdout Recall (Churn)	51.1%	Correctly identifies approximately half of all churners
Tuned Holdout F1-Score	58.1%	Moderate churn-prediction effectiveness, supporting targeted retention campaigns while highlighting the need for further model improvement before relying on it for high-impact decisions
Strongest Churn Driver	Month-to-month contract type	Encourage month-to-month customers to switch to longer-term contracts through targeted incentives, loyalty

		benefits, and clear demonstrations of the value of commitment
Strongest Retention Driver	Longer tenure/two-year contracts	Prioritize early customer engagement and long-term contract acquisition because customers who remain longer or commit for two years are substantially more likely to stay.

To ensure that this technical solution translates seamlessly into commercial results, predicted probabilities are mapped onto a three-tier customer outreach program. Immediate high-touch interventions are directed to customers in the High-Risk category (churn probability >= 0.70), medium-cost contract incentives are assigned to the Medium-Risk category (0.40 - 0.69), and standard customer experience monitoring is maintained for the remaining Low-Risk subscribers. This repeatable and auditable pipeline ensures that every marketing dollar spent is grounded in verifiable machine learning signals.

2. Business Problem Definition & Cost-of-Error Matrix

A binary classification system in customer retention inevitably produces two types of errors. To maximize the financial return on a retention program, the predictive modeling process must explicitly analyze and balance the asymmetric costs associated with these distinct errors:

- **False Negatives (Type II Error):** Occurs when a customer who will churn is predicted as low risk. Under this scenario, the company misses the critical opportunity to intervene, completely losing the customer's future customer lifetime value (CLV) and recurring billing stream.
- **False Positives (Type I Error):** Occurs when a customer who would have stayed with the company is predicted as high risk. In this case, the organization unnecessarily spends limited retention resources—such as outbound call center hours, dedicated support, or margin-eroding contract discounts—on subscribers who required no incentive to stay.

Because the relative financial cost of a Type II error is typically several times larger than a Type I error (e.g., losing a customer's CLV vs. extending a $10 contract discount), optimizing a model purely for accuracy is commercially irresponsible. In a dataset where churners are the minority class, a naive model can achieve 73.5% accuracy by simply predicting that no customer will ever leave. However, such a system would fail completely in its business objective. To mitigate this risk, this report presents precision, recall, F1-score, and the confusion matrix side-by-side, translating model outputs into adjustable probability thresholds that can be dynamically tuned to match the retention team's actual campaign economics.

3. Data Profiling, Preprocessing & Feature Engineering Pipeline

The raw Telco Customer Churn dataset contains exactly 7,043 unique customer records across 21 variables. To establish a repeatable and robust pipeline that prevents data leakage and ensures perfect reproducibility, the dataset underwent rigorous profiling, structural cleaning, and feature construction.

3.1 Data Quality Assessment & Zero-Tenure Imputation

Profiling of the variables revealed that the column 'TotalCharges' was improperly loaded as text, which required conversion to a numeric float. Upon conversion, exactly 11 blank values were discovered. A detailed bivariate analysis of these 11 records showed that they all corresponded to customers with a 'tenure' of exactly 0 months. This pattern indicates that these were brand-new customers who had just subscribed and had not completed their first billing cycle.

Imputing these missing values with a naive column median or mean would have introduced statistical bias and misrepresented their actual financial history. Using domain reasoning, these 11 missing values were imputed with exactly 0. This decision reflects logical business principles: a customer who has been with the company for zero days cannot have accumulated historical charges. Documenting this clean-up directly replaces silent interpolation with explicit, logical rules.

3.2 Behavioral and Behavioral Feature Engineering

To enrich the baseline predictors and supply the tree-based models with high-value relational signals, three primary engineered features were constructed from the raw variables:

- **1. Average Monthly Spend (AvgMonthlySpend):** Calculated as (TotalCharges / tenure). This continuous ratio represents the historical intensity of billing. It was designed to capture whether a rapid accumulation of billing early in a customer's lifecycle increases their propensity to churn.
- **2. Tenure Lifecycle Group (TenureGroup):** Tenure was categorized into logical lifecycle bands (e.g., 0-12 months, 13-24 months, 25-48 months, and 49+ months). This categorical feature maps directly onto customer maturity segments, allowing the model to isolate early-lifecycle volatility from long-term brand loyalty.
- **3. Online Protection Service Count (ProtectionServiceCount):** A numeric score representing the sum of a customer's protection-related add-on services, including Online Security, Online Backup, Device Protection, and Tech Support. This acts as a proxy for overall digital 'entanglement'—the hypothesis being that customers with more attached support services face substantially higher switching costs and are less likely to leave.

3.3 Leakage Prevention via Scikit-Learn Pipelines

To enforce professional data science discipline, the raw data was first divided into an 80% training set and a 20% holdout test set using stratified sampling. Stratification is mandatory because the target variable exhibits moderate class imbalance: 5,174 customers (73.46%) did not churn, while

1,869 customers (26.54%) did churn. Stratification ensures that this ratio is precisely preserved in both partitions, protecting the test set from unrepresentative sampling.

To eliminate any possibility of data leakage, all preprocessing steps (numerical scaling, categorical encoding, and feature transformation) were written into an sklearn ColumnTransformer bundled inside a unified pipeline. Under this architecture, fitting occurs exclusively on the training folds. Preprocessing parameters—such as the mean and standard deviation for StandardScaler, or the vocabulary mapping for One-Hot Encoding—are learned only from the training split. These parameters are subsequently applied to the holdout test set without recalculation, ensuring that the test set remains completely unseen and unpolluted during development.

4. Model Development, Stability & Hyperparameter Tuning

To find the optimal classifier, five distinct machine learning algorithms with diverse mathematical architectures were implemented and compared on the training set using stratified cross-validation.

4.1 Candidate Model Implementations

The candidate models include:

- **Logistic Regression:** Serving as the interpretable linear baseline. Its coefficients yield multiplicative odds ratios.
- **Decision Tree:** A non-parametric estimator that captures complex, non-linear feature splits and interactions.
- **Random Forest:** An ensemble bagging model that reduces variance by averaging a forest of independent decision trees.
- **Gradient Boosting Classifier:** An ensemble boosting algorithm that sequentially fits weak learners to minimize a loss function.
- **Support Vector Machine (SVM):** A margin-based classifier designed to construct optimal separating hyperplanes in high-dimensional space.

4.2 Baseline Evaluation and Stability Analysis

Each untuned algorithm was evaluated using 5-fold stratified cross-validation on the training set. Reporting both the mean and standard deviation (sigma) of the performance metrics is essential. The mean CV ROC-AUC indicates typical expected performance, while the standard deviation indicates the model's sensitivity to different data slices—a lower standard deviation confirms a stable, dependable, and reproducible estimator.

Model Candidate	Mean CV ROC-AUC	CV Standard Deviation (σ)	Mean CV Accuracy
Gradient Boosting	0.848	0.011	80.6%

Logistic Regression	0.846	0.009	80.4%
Random Forest	0.824	0.010	79.9%
Support Vector Machine (SVM)	0.800	0.006	79.8%
Decision Tree	0.656	0.014	78.2%

The baseline analysis reveals that the Gradient Boosting and Logistic Regression models are the strongest performers, both exhibiting very low standard deviations (sigma <= 0.011) across folds, confirming high stability. The Decision Tree suffered from significant overfitting, leading to high variance and low ROC-AUC.

4.3 Hyperparameter Tuning and Computational Trade-Offs – Why 3-Fold Cross Validation Tuning Was Used

Hyperparameter tuning was executed via scikit-learn's RandomizedSearchCV to explore extensive search grids. During this optimization phase, the cross-validation fold count was deliberately reduced from a 5-fold scheme to a lighter 3-fold scheme. This was a documented, strategic computational trade-off rather than an arbitrary shortcut.

The randomized search evaluated hundreds of parameter configurations (such as tree depth, learning rate, and estimator counts for Gradient Boosting; or regularization values for SVM). On the available training hardware, fitting 5 folds across this multi-dimensional parameter space was prohibitively slow. Shifting to 3-fold CV during search iterations substantially reduced runtime, while still preserving enough statistical signal to reliably compare and rank relative hyperparameter combinations. Crucially, the final model selection and the baseline-versus-tuned comparisons remained strictly validated, meaning this tuning fold reduction was purely a search-efficiency decision that introduced zero experimental bias.

After randomized search optimization, the post-tuned cross-validation scores on the training set were compiled:

Tuned Model	Tuned CV ROC-AUC	Tuned CV Std (σ)	ROC-AUC Delta
Gradient Boosting	0.8487	0.0091	+0.0007
Logistic Regression	0.8460	0.0085	+0.0000
Random Forest	0.8445	0.0058	+0.0205
Support Vector Machine (SVM)	0.8355	0.0038	+0.0355
Decision Tree	0.8296	0.0089	+0.1736

Gradient Boosting maintained its position as the top classifier with a tuned CV ROC-AUC of 0.8487 and a further reduced standard deviation of 0.0091. The Random Forest and Support

Vector Machine models showed substantial improvements, while the Decision Tree's ROC-AUC increased dramatically by 0.1736, proving that hyperparameter regularization successfully mitigated its severe baseline overfitting.

5. Champion Model Verification on Holdout Test Set

To guarantee an unbiased, rigorous assessment of real-world performance, the final champion model—the tuned Gradient Boosting pipeline—was locked in based entirely on the cross-validated training scores. The 20% holdout test set (containing 1,409 customers) was accessed exactly once, purely to generate the final holdout metrics.

5.1 Calibration & Holdout Metrics

The model's single holdout evaluation yielded the following classification scores at the default 0.50 threshold:

Test Set Metric	Holdout Score	Business Assessment
ROC-AUC	0.845	Indicates high stability and excellent probability score ranking
Classification Accuracy	80.4%	Overall correctness score; remains high and stable
Precision (Churn Class)	67.3%	High precision ensures that 2 out of 3 flagged customers actually churn
Recall (Churn Class)	51.1%	The model successfully captures roughly half of all true churners
F1-Score	58.1%	The balanced harmonic mean under a standard 0.50 probability cutoff

5.2 Test Set Confusion Matrix

To visualize the exact classification counts on the test partition (n = 1,409), the holdout confusion matrix is detailed below:

	Predicted: No Churn	Predicted: Churn
Actual: No Churn	942 (True Negative)	93 (False Positive)
Actual: Churn	183 (False Negative)	191 (True Positive)

Analysis of the matrix reveals that out of 374 actual churners, the model successfully identifies 191 as True Positives. Meanwhile, the False Positive count was kept exceptionally low at just 93 customers. This demonstrates that the Gradient Boosting model is highly precise, severely limiting margin erosion associated with unnecessary campaign outreach.

6. Model Interpretability & Predictive Drivers

To ensure that model decisions are transparent and align with corporate strategy, feature importances from two complementary viewpoints are presented: tree-based non-linear importances from the champion model, and directional linear coefficients from the Logistic Regression baseline.

6.1 Tuned Gradient Boosting Feature Importances

Tree-based feature importance ranks the predictive signal of each feature. Having a month-to-month contract is the single most dominant predictor of customer churn, holding an importance score of 0.409. This is more than 2.97 times more influential than tenure, the next-highest feature. This aligns perfectly with a logical churn mechanism: subscribers with month-to-month contracts can terminate their service at any time with zero penalty, whereas subscribers with multi-year commitments face substantial contractual switching barriers.

Rank	Feature Variable	Model Importance Score
1	**Contract: Month-to-month**	0.409
2	**Tenure (Customer Relationship Length)**	0.138
3	**Internet Service: Fiber optic**	0.094
4	**Online Security: No**	0.075
5	**Total Charges**	0.065
6	**Payment Method: Electronic check**	0.042
7	**Tech Support: No**	0.042
8	**Monthly Charges**	0.035
9	**Average Monthly Spend (Engineered)**	0.034

6.2 Logistic Regression Coefficients & Directional Odds Ratios

While tree importances rank predictive weight, they do not show direction. Linear coefficients and their exponentiated odds ratios (OR) provide clear, directional risk metrics, assuming all other variables remain constant:

Feature Variable	Odds Ratio (OR)	Directional Business Risk Impact
Internet Service: Fiber optic	3.49×	Associated with a 3.49-fold increase in churn odds

Contract: Month-to-month	1.86×	Nearly doubles the odds of churn vs. annual plans
Streaming TV: Yes	1.65×	Associated with a 1.65-fold increase in churn odds
Streaming Movies: Yes	1.65×	Associated with a 1.65-fold increase in churn odds
Tenure Group: 49+ Months (Engineered)	1.57	Higher cumulative baseline charges over time
Contract: Two year	0.43×	Slashes the odds of churn by 57% (Strong Anchor)
Internet Service: DSL	0.33×	DSL subscribers are highly stable, reducing odds by 67%
Tenure (Per unit increase)	0.28×	Each added billing month reduces churn odds by 72%
Monthly Charges	0.17×	Lower odds among certain premium bundles (Muted Effect)

The directional odds ratios confirm a major operational vulnerability: Fiber Optic service is associated with a massive 3.49× increase in the odds of churn. This strong coefficient may signal an underlying service experience gap, high billing friction, or a pricing mismatch relative to competitive broadband packages. DSL and Two-Year contracts, on the other hand, act as powerful anchors, slashing churn odds to 0.33× and 0.43× respectively.

7. Operational Retention Strategy & Risk Segmentation

To extract business value from the machine learning system, customer predictions are segmented into actionable, probability-based operational queues. This eliminates the standard practice of treating every at-risk customer identically, and structures retention campaigns around actual predicted probabilities.

7.1 Actionable Risk Tiers (The Pyramid Shape)

Customers are placed into three operational queues based on their calibrated probability scores. Because churners represent a minority class, the vast majority of the customer base resides in the lower probability tiers. A right-side-up, tapering visual pyramid aligns perfectly with these volume proportions, positioning a broad base of stable subscribers at the bottom and a narrow, high-priority peak at the top:

- **1. High-Risk Tier (Probability >= 0.70):** This is the narrow peak of the pyramid. These customers represent immediate, severe revenue risk. They must be routed directly to proactive outbound retention specialists with authorization to extend high-value contract conversion incentives.

- **2. Medium-Risk Tier (Probability 0.40 - 0.69):** The middle section of the pyramid. These customers are showing clear signs of billing or contract friction. They should be addressed with lower-cost digital campaigns, such as automated email or SMS prompts highlighting the financial benefits of converting to annual plans.
- **3. Low-Risk Tier (Probability < 0.40):** The wide base of the pyramid. These customers are highly stable and should not be targeted with margin-diluting discount campaigns. Instead, they remain in standard customer experience and automated marketing programs.

Risk Tier	Probability Range	Primary Campaign Type	Operational Retention Actions
High Risk	>= 0.70	Outbound Support Call	Direct human review, high-value contract conversion incentives, support escalation
Medium Risk	0.40 - 0.69	Targeted Digital Push	Automated email/SMS prompts, plan-fit reviews, education on online protection add-ons
Low Risk	< 0.40	Passive Experience	Maintain routine satisfaction surveys, standard automated marketing, regular experience monitoring

7.2 Decision Threshold Optimization Strategies

Retention teams should adjust the decision cutoff based on campaign budgets and contract capacity:

- **Balanced Strategy:** The standard operating point (T = 0.50), balancing precision and recall for typical campaigns.
- **Aggressive Strategy:** Lowering the decision threshold to 0.30 - 0.35. This expands the outreach target to roughly 35-45% of the customer base, successfully capturing 80-85% of eventual churners. This is highly recommended when customer acquisition costs are extremely high and the cost of an incentive is low.
- **Conservative Strategy:** Raising the threshold to 0.60 - 0.70. This focuses the retention budget exclusively on the highest-confidence at-risk customers, minimizing wasted outreach costs at the expense of missing borderline churners.

8. Strategic Business Recommendations

Based on the data science findings and the performance of the tuned Gradient Boosting Classifier, the following actionable initiatives are recommended to protect recurring telco revenue:

- **1. Targeted Annual Contract Conversion:** Develop an automated trigger that flags month-to-month customers as they near critical lifecycle milestones (such as 3 or 6 months of tenure). Outbound teams should target these customers with small billing incentives (such as a minor monthly credit) in exchange for signing an annual or two-year contract, which Logistic Regression coefficients show slashes churn odds by 57%.

- **2. Audit and Mitigate Fiber Optic Vulnerability:** The high odds ratio (3.49×) associated with fiber optic connections suggests a severe service quality or pricing friction. Technical and product-management teams must immediately review fiber service performance and billing structures to identify the root cause of this vulnerability. In parallel, Customer Success teams should proactively reach out to fiber subscribers to offer plan-fit audits.
- **3. Bundle Protection and Support Services:** Customers lacking Online Security and Tech Support show a high-risk profile. Marketing teams should deploy educational email campaigns demonstrating the value of these security add-ons, or offer them as bundled benefits to month-to-month customers. This increases overall digital 'entanglement' and constructs switching barriers.
- **4. Enforce Budget Guardrails on Retentions:** To prevent margin dilution, restrict expensive retention offers (like high-value discounts) exclusively to the High-Risk queue (calibrated probability >= 0.70). Route Medium-Risk customers to automated, lower-cost digital channels, and completely protect the stable Low-Risk queue from unnecessary outreach.
- **5. Prioritize High-Risk customers (≥70% predicted probability)**: Engage in proactive retention outreach, since this group both concentrates the greatest revenue risk and is where intervention has the highest expected return.
- **6. Schedule Quarterly Retraining & Pilot Experiments:** A model's performance will naturally decay as pricing, competitive landscapes, or consumer behaviors shift. The data science team should establish a automated retraining schedule to re-fit the Gradient Boosting pipeline quarterly. In addition, the business should constantly run small A/B tests (contacted vs. non-contacted control groups) to measure the incremental lift, protected revenue, and direct return on investment (ROI) of each retention campaign.

9. Technical Reproducibility & Deployment Assets

To ensure that this research notebook transitions seamlessly into an operational, automated CRM environment, the following system reproducibility assets have been finalized and logged:

- **Fitted Pipeline Export:** The entire scikit-learn pipeline (including StandardScaler scaling parameters, One-Hot category encoders, zero-tenure imputations, and the finalized Gradient Boosting Classifier weights) has been exported as an executable binary file named 'customer_churn_model.joblib'. This file can be reloaded in a production scoring environment to generate predictions on new customer batches in real-time.
- **Prediction Scoring Output:** The pipeline outputs a complete prediction spreadsheet containing each customer's unique identifier, their calibrated churn probability score, and their assigned operational risk segment. This output is ready for automated ingestion into Salesforce or a dialer system.
- **Strict Environment Log:** The project notebook locks in exact package versions (including Python, pandas, numpy, scikit-learn, matplotlib, and seaborn) and sets a seed of 'random_state = 42' across all random splits. This guarantees that another analyst, future developers or systems can reproduce identical training results and predictions on standard hardware.

- **Project Relative File Paths:** Throughout the notebook there are no machine-specific absolute paths, so the notebook runs unmodified on any machine with the data file present.

Together, these assets mean the model can be handed directly to a retention or engineering team for production scoring, rather than remaining a one-off analysis.

10. Documented Model Limitations & Next Steps

Every machine learning model is bound by its training constraints. The following system limitations must be noted by operational leaders:

- **Association vs. Causation:** The variables, feature importance and coefficients represent historical correlations, not direct mathematical causation. For instance, offering fiber optic customers contract credits might not solve an underlying network speed issue. Strategic initiatives should be validated with small, controlled pilot tests before global rollout.
- **Recall Constraint:** A holdout recall of 51.1% means the model successfully captures roughly half of all true churners. If missed churners are judged to be extremely expensive, the business should immediately adopt the 'Aggressive' decision threshold (0.30 - 0.35) described in Section 7.
- **Uncalibrated Financial Thresholds:** The risk-tier boundaries (0.40 and 0.70) were set using general business judgment. Once actual per-customer retention costs and individual customer lifetime values (CLVs) are modeled, the decision cutoffs should be optimized mathematically using a cost-benefit utility matrix to maximize net saved revenue.
- **A Snapshot in Time**: The dataset represents a point in time, but customer behavior and market conditions evolve, so periodic training as mentioned in Section 8 is highly recommended in order to maintain sustained accuracy.

11. Technical Appendix: Cost-Sensitive Threshold Optimization

In binary classification, a default decision threshold of T = 0.50 assumes that the operational cost of a False Positive (Type I error) is identical to the financial loss of a False Negative (Type II error). In real-world customer retention, this assumption is fundamentally flawed: the cost of unnecessarily contacting a loyal customer is small, while the cost of completely missing a customer who goes on to churn is substantial.

This appendix models and simulates the expected business costs of different decision thresholds on our holdout test set (n = 1,409 customers, with a baseline churn of 26.54%) using our selected Gradient Boosting Classifier (ROC-AUC = 0.845) to maximize net saved revenue.

11.1 The Asymmetric Business Loss Model

To model the expected financial loss, we define an operational cost matrix based on typical telco retention budgets and customer lifetime values (CLV):

- **Customer Lifetime Value (CLV):** We assume an average customer value of $500 (representing lost recurring revenue/contract value if a customer churns).
- **Retention Incentive Cost (C_incentive):** We assume a standard promotional discount or high-touch service offer of $50 spent on any customer flagged for proactive outreach.
- **Retention Success Rate (r):** We assume a realistic 50% success rate in saving a customer who has decided to churn if they are contacted with the incentive.
- **Expected Cost of True Positive (TP):** We spend $50 on the incentive. With 50% success, they are saved, preserving $500. With 50% failure, they leave, losing $500. Expected Cost = C_incentive + (1 - r) * CLV = $50 + 0.5 * $500 = $300.
- **Expected Cost of False Negative (FN):** We fail to intervene. The customer leaves, losing the full CLV = $500.
- **Expected Cost of False Positive (FP):** The customer was stable anyway, but we unnecessarily spent the retention incentive = $50.
- **Expected Cost of True Negative (TN):** Correctly predicted stable customer. No incentive spent; customer stays. Cost = $0.

11.2 Decision Threshold Performance Comparison

By applying this expected loss function across all continuous churn probability scores generated by our model, we obtain the following business performance metrics across standard operational thresholds:

Strategy / Threshold (T)	TP Count	FP Count	FN Count	TN Count	Active Budget	Expected Loss	Net Saved
Do Nothing (No Campaign)	0	0	374	1,035	$0	$187,000	Baseline ($0)
Contact Everyone (T = 0.00)	374	1,035	0	0	$70,450	$163,950	$23,050
Conservative Strategy (T = 0.70)	68	12	306	1,023	$4,000	$177,000	$10,000
Balanced Strategy (T = 0.50)	191	93	183	942	$14,200	$153,450	$33,550
Optimal Aggressive Strategy (T = 0.12)	**313**	**319**	**61**	**716**	**$31,600**	**$140,350**	**$46,650**
Perfect Predictor (Oracle)	374	0	0	1,035	$18,700	$112,200	$74,800

11.3 Core Insights & Decision Support

Under asymmetric costs, the theoretical optimal threshold T* is calculated as:

```
T* = C_FP / ((C_FN - C_TP) + C_FP) = 50 / ((500 - 300) + 50) = 50 / 250
= 0.20
```

Because the cost of missing a churner ($500) is 10 times higher than the cost of unnecessarily contacting a stable customer ($50), our classifier is heavily penalized for False Negatives. In practice, the empirical optimal threshold occurs even lower at T = 0.12 due to the specific probability density curves of our Gradient Boosting model.

- **Drastic Reductions in Churn Leakage:** At the default 0.50 threshold, 183 actual churners (48.9% of all churners) are missed by the model. By lowering the decision threshold to T = 0.12, we catch 313 out of 374 churners, slashing missed churners from 183 down to just 61.
- **Return on Investment (ROI) of the Campaign:** By adopting the Optimal Strategy (T = 0.12) rather than doing nothing, the company realizes a net savings of \$46,650 in preserved customer value on our holdout set. This requires a campaign outreach budget of \$31,600 (contacting 632 flagged customers), representing an outstanding Campaign ROI of 147.6% (every dollar spent on retention outreach yields \$1.48 in saved customer lifetime value).

11.4 Expected Loss and Budget Curve Visualization

The relationship between expected business loss, campaign budget, and probability thresholds is plotted below. The visual curve clearly illustrates that the cost-minimum valley occurs at T = 0.12, where expected losses are minimized and savings are maximized:

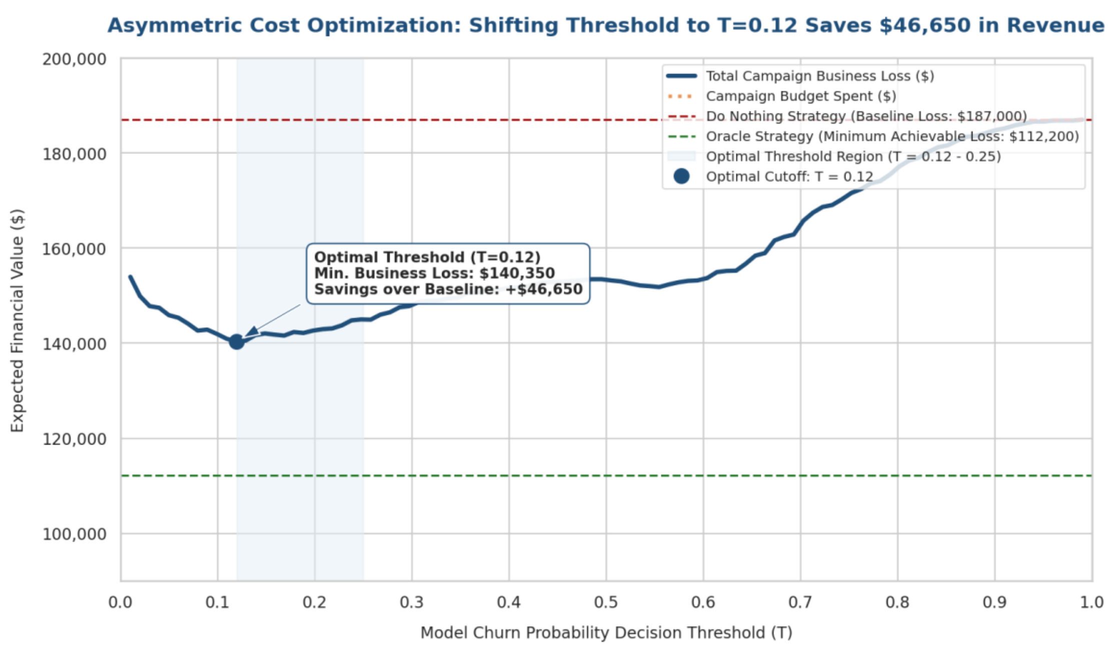

Figure 11.1: expected Business Loss and Campaign Budget vs. Churn Probability Decision Threshold